COSC264 · INTRODUCTION TO COMPUTER NETWORKS AND THE INTERNET · TERM 4

Module 3
Error Detection and Correction

cosc264.up.railway.app

MODULE 3

Learning objectives & module plan

CONCEPT 1
Error Detection

CONCEPT 2
Cyclic Redundancy Check

CONCEPT 3
Error Correction

We introduce **error detection** and two fundamental schemes: **parity check** and the **Internet checksum**.

We then talk about the **Cyclic Redundancy Check (CRC)**, which is a more powerful error-detection scheme.

We end with **error correction**: not only detecting that something went wrong, but repairing it.

MODULE 3 · CONCEPT 1 OF 3

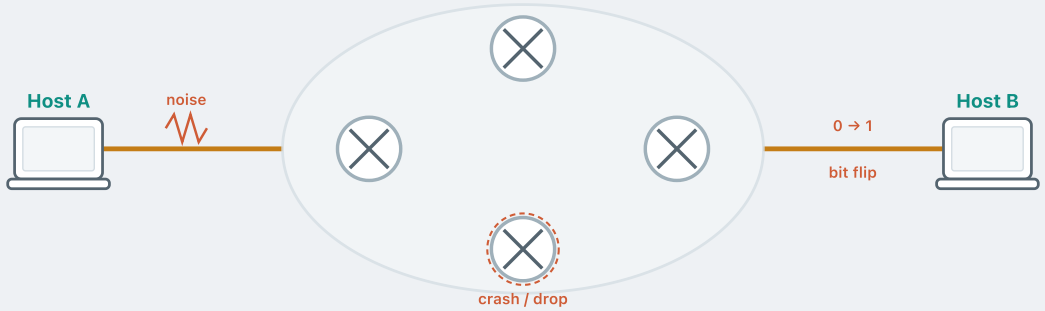
Error Detection

1 · Error Detection | 2 · CRC | 3 · Error Correction

MOTIVATION

The journey of a packet

Consider sending a datagram from **host A** to **host B** across a network of links and routers.



Along the way — **thermal noise**, weak signals, interference, equipment crashes — the packet might arrive **corrupted**, arrive **out of order**, or be **dropped** entirely.

MOTIVATION

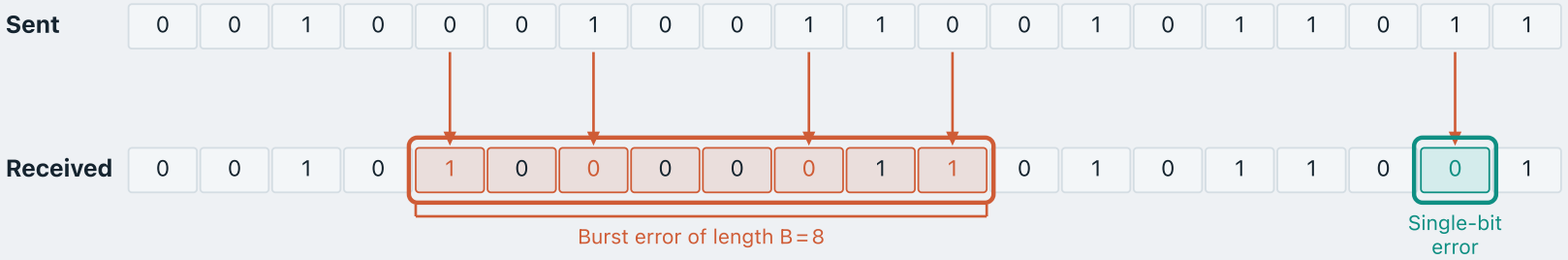
Three categories of problem

PROBLEM	APPROACH
Corrupted packets this module	Error detection and correction
Lost packets (Module 4)	ARQ protocols
Out-of-order packets (Module 4)	ARQ protocols

> **DEFINITION**
> A packet is **corrupted** when it is delivered, but one or more of its bits have been **altered in transit.** Those altered bits are called **bit errors.**

ERROR TYPES

Types of bit errors



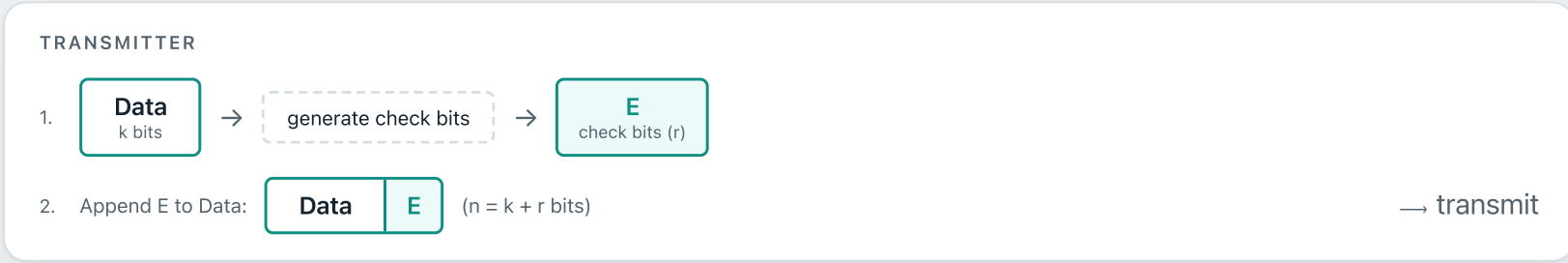
Burst error of length B — the span from the first to the last corrupted bit is B positions.

Single-bit error — a single isolated bit is altered.

ERROR DETECTION PROCESS — SENDING

The error detection process

The sender runs the data through an **error-detecting code**, which produces a small number of **check bits** *E* — a compact fingerprint that depends on every bit of the original content.



ERROR DETECTION PROCESS — RECEIVING

The error detection process

RECEIVER

1. Receive and parse: **Data′** | **E** → **Data′** (k bits) + **E** (check bits (r))
2. **Data′** → recompute check bits → **E′** (check bits (r))
3. Compare **E** (received) with **E′** (recomputed)

E = E′ → no error detected — accept the data.

E ≠ E′ → error detected — discard or request retransmission.

- If nothing is corrupted, the check **always** passes.
- The reverse is **not** true. A check that passes does not guarantee nothing was corrupted — it only says **no error was detected**. Some rare corruption patterns still slip through.
- So a **good** error-detecting code should catch **as many error patterns as it can**: if an error occurred, it is detected with **high probability**.

PARITY CHECK

Parity check

The simplest **error-detecting** scheme: append a **parity bit** to each block of data.

Even parity — parity bit makes the total number of 1s (including the parity bit) even.

TRANSMITTER

1. `1 0 0 0 0 1 0 1` data (3 ones) → even parity → **1** parity bit — 3 + 1 = 4 (even) ✓
2. Append: `1 0 0 0 0 1 0 1` | **1** → ⟶ transmit

RECEIVER — bit 4 flipped in transit

1. Parse: `1 0 0 1 0 1 0 1` data′ (4 ones) + **1** received
2. `1 0 0 1 0 1 0 1` → even parity → **0** recomputed
3. Compare: **1** (received) ≠ **0** (recomputed) → **Error detected!**

Odd parity is the same idea with the opposite convention — the parity bit makes the total number of 1s **odd**.

PARITY CHECK

Parity check — can it be fooled?

Can you think of a scenario where the parity check **fails** — the data arrives corrupted, but the receiver sees nothing wrong?

Take the same even-parity example, and let **two** bits flip — one of them the parity bit itself.

IN TRANSIT — 2 BITS FLIPPED

`10000101 | 1` → `10010101 | 0` bit 4: 0→1, parity bit: 1→0

RECEIVER

1. Parse: `10010101` data′ (4 ones) + `0` received
2. `10010101` → even parity → `0` recomputed
3. Compare: **0** = **0** → **No error detected!** (but the data is corrupted)

A parity check cannot detect a corruption pattern in which an **even number of bits** are flipped.

We need a more **robust** error-detection scheme.

2-dimensional parity check

Arrange the data into a **matrix** of rows and columns.

For example, a 20-bit data block becomes the 4 × 5 matrix on the right.

Append a **row parity bit** r_i to each row (even parity).

Append a **column parity bit** c_j to each column.

An **overall parity bit** p completes the matrix.

0	1	1	1	0	**1**
0	1	1	1	0	**1**
0	1	0	0	0	**1**
0	1	0	1	1	**1**
0	**0**	**0**	**1**	**1**	0

↑ column parity row parity →

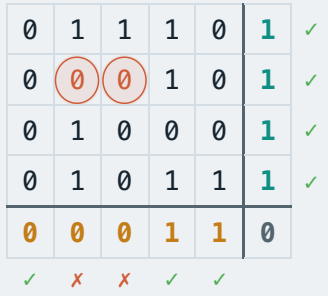
2-D PARITY

How much can 2-D parity catch?

Flip **two bits in the same row** — the pattern that defeated the one-dimensional check.

The **row** check still passes: within that row, one flip cancels the other.

But each flip is alone in its **column**, so it flips that column's parity — the **column check catches the error**.



In general, every pattern of **one, two or three** flipped bits is detected.

Can you think of an error pattern that 2-D parity would miss?

2-D PARITY

How much can 2-D parity catch?

Flip **two bits in the same row** — the pattern that defeated the one-dimensional check.

The **row** check still passes: within that row, one flip cancels the other.

But each flip is alone in its **column**, so it flips that column's parity — the **column check catches the error**.

> In general, every pattern of **one, two or three** flipped bits is detected.

> Can you think of an error pattern that 2-D parity would miss?

Consider the four flipped bits on the right, at the corners of a **rectangle** — every check **passes**, and the error goes **undetected**.

2-D PARITY

Correcting a single-bit error

2-D parity can not only **detect** an error — if just a single bit was flipped, it can also **correct** it.

The matrix on the right is as transmitted.

0	1	1	1	0	1
0	1	1	1	0	1
0	1	0	0	0	1
0	1	0	1	1	1
0	0	0	1	1	0

Correcting a single-bit error

2-D parity can not only **detect** an error — if just a single bit was flipped, it can also **correct** it.

The matrix on the right is as transmitted.

During transmission, a single bit is **flipped**.

The receiver's parity check **fails for row 2** and **column 2**.

0	1	1	1	0	1	✓
0	0	1	1	0	1	✗
0	1	0	0	0	1	✓
0	1	0	1	1	1	✓
0	0	0	1	1	0	
✓	✗	✓	✓	✓		

Correcting a single-bit error

2-D parity can not only **detect** an error — if just a single bit was flipped, it can also **correct** it.

The matrix on the right is as transmitted.

During transmission, a single bit is **flipped**.

The receiver's parity check **fails for row 2** and **column 2**.

0	1	1	1	0	1	✓
0	1	1	1	0	1	✗
0	1	0	0	0	1	✓
0	1	0	1	1	1	✓
0	0	0	1	1	0	
✓	✗	✓	✓	✓		

> The failing row and column **intersect** at the corrupted bit — flip it to **correct** the error.

Internet checksum

The **Internet checksum** is a specific **error-detecting code** used in **IPv4**, **TCP**, and **UDP** (discussed later) to let the receiver detect bit errors introduced during transmission.

IPv4 HEADER — 32 bits

Version	IHL	Diff. Services	Total Length	
Identification			Flags	Fragment Offset
TTL		Protocol	**Header Checksum**	
Source Address				
Destination Address				
Options (0 or more words)				

TCP HEADER — 32 bits

Source Port			Destination Port
Sequence Number			
Acknowledgement Number			
Data Offset	Rsv	Flags	Window Size
Checksum			Urgent Pointer
Options (0 or more 32-bit words)			
Data (optional)			

One’s complement arithmetic

The checksum uses two operations:

One’s complement addition

1. Add the two numbers as unsigned integers.
2. If there is a carry out of the leftmost bit, add 1 to the sum (*end-around carry*).

```
ex1.  (no carry)
  A2B3
+ 1D4C
------
  BFFF
```

```
ex2.  (carry out)
      A2B3
    + 6F8E
    ------
carry → 1 1241
    +     1 ← end-around
    ------
      1242
```

One’s complement operation

Replace every 0 bit with 1 and every 1 bit with 0.

```
1111 → ones complement → 0000
1001 → ones complement → 0110
```

Adding a value and its one’s complement always gives all 1s:

```
1001 + 0110 = 1111
```

INTERNET CHECKSUM

Sender computation

1. Take the **IP header**.
2. Split it into **16-bit words**.
3. Compute their **one's-complement sum**.
4. Take the **one's complement** of that sum — that is the **checksum**.

IPv4 header rows are 32 bits, so each can be written as **8 hex digits**:

`F2345678`

`F234` `5678` 32 bits → two 16-bit words

	HEX	BINARY
Word 1	F234	1111 0010 0011 0100
Word 2	5678	0101 0110 0111 1000
Add	148AC	1 0100 1000 1010 1100
Wrap carry	+ 1	+ 1
One's-complement sum	48AD	0100 1000 1010 1101
Checksum = complement	**B752**	1011 0111 0101 0010

Receiver verification

1. Take the received header — the checksum field now holds **B752**.
2. Compute the **one's-complement sum** of the header words.
3. Add the **checksum** to that sum.
4. The checksum is the **one's complement** of that sum, so adding them must give **all 1 bits** — check that the total is `FFFF`.

`F234 5678 B752`

	HEX	BINARY
Word 1	F234	1111 0010 0011 0100
Word 2	5678	0101 0110 0111 1000
Add words 1–2	148AC	1 0100 1000 1010 1100
Wrap carry	+ 1	+ 1
One's-complement sum	48AD	0100 1000 1010 1101
Add checksum	+ B752	+ 1011 0111 0101 0010
Total	**FFFF**	1111 1111 1111 1111

Efficiency and limitations

The primary reason for its adoption is **efficiency**:

- Most of these protocols are implemented in software
- The Internet checksum involves simple operations → very little computational overhead

But it also has a **limitation**:

- It is much less effective at detecting errors than the **cyclic redundancy check (CRC)**, which we look at next

MODULE 3 · CONCEPT 2 OF 3

Cyclic Redundancy Check

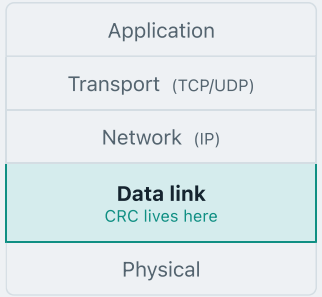
1 · Error Detection | **2 · CRC** | 3 · Error Correction

ERROR DETECTION

Cyclic Redundancy Check (CRC)

The CRC is one of the **most common and most powerful** error-detecting codes in use.

It lives in the **data link layer**. Every frame you send over Ethernet or Wi-Fi carries a CRC.



> **The intuition:** sender and receiver agree on a fixed **divisor** in advance. Both read the message as a **number**, and the sender appends a few check bits that make the whole thing **exactly divisible** by that divisor.

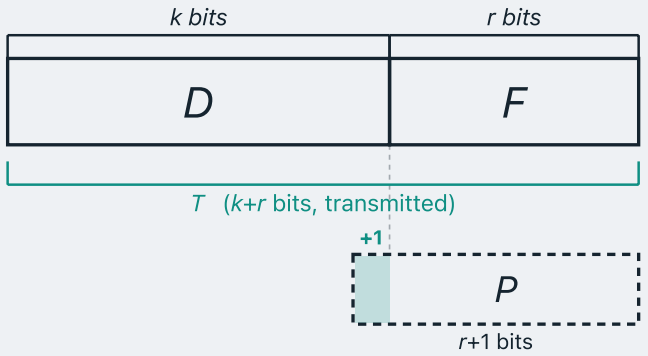
CRC — notation

D — the *k*-bit block of data to send

P — the **divisor**, a fixed pattern of *r*+1 bits that sender and receiver agree on in advance

F — the **frame check sequence (FCS)** we must compute, *r* bits long

T — the frame actually transmitted, *k*+*r* bits: *D* followed by *F*



Some texts write *F*'s length as *n*−*k*, where *n* = *k*+*r* is the whole frame.

How to choose the pattern P?

Not every pattern will do — two **clearly bad** choices:

- **P = 0000…0** — you cannot divide by zero at all.
- **Highest-order bit of P = 0** — the leading zero contributes nothing to the division, so P is effectively **one bit shorter**. And since the FCS is always one bit shorter than P, the **effective FCS is one bit shorter** too — fewer check bits than we intended.

  Intuitively, fewer check bits means weaker error detection — not what we want.

The exact bit pattern depends on the **type of errors** expected — but in every case:

REQUIRED

The **highest-order bit** = 1

REQUIRED

The **lowest-order bit** = 1

This is what buys the **burst-error guarantee** — we will see why shortly.

Modulo-2 Arithmetic

- Addition and subtraction are both performed with **XOR (⊕)** — so they are the **same operation**.

ADDITION (XOR)

```
  1111
⊕ 1010
------
  0101
```

SUBTRACTION (XOR)

```
  1111
⊕ 1010
------
  0101
```

- Multiplication is the usual shift-and-add — only the additions inside it become XORs.

MULTIPLICATION

```
    11001
×     11
-------
    11001
⊕ 110010
---------
   101011
```

How to compute the FCS

GIVEN

D **1010001101** *k* = 10 bits

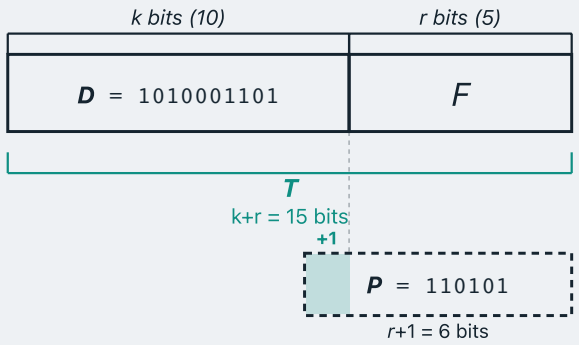
P **110101** *r*+1 = 6 bits

TO FIND

F, the **FCS** — one bit shorter than *P*, so *r* = **5 bits** — such that:

1. appending *F* to *D* gives *T*, of *k*+*r* = 10 + 5 = **15 bits**;
2. as integers, $T = D \times 2^r + F$;
3. and *T* is **exactly divisible by *P***.

$$F = \text{remainder of } (D \times 2^r) / P$$



Plugging in the numbers:

$$F = \text{rem. of } (1010001101 \times 2^5) / 110101$$
$$= \text{rem. of } 101000110100000 / 110101$$

An Example — Computing FCS

```
              1101010110   quotient (discarded)
          ---------------
110101 ) 101000110100000   D × 2⁵
         110101000000000   ⊕ P
         ---------------
          11101110100000
          11010100000000   ⊕ P shifted
          ---------------
            111010100000
                ⋮  four more steps  ⋮
                  -----
                  01110    remainder
```

We need the remainder of `101000110100000 ÷ 110101`, and we get it by **long division** — the same procedure you learned at school.

The only difference: at each step the subtraction is an **XOR**, because we are working in **modulo-2 arithmetic**.

***F* = 01110** (the 5-bit FCS)

Transmitted frame *T*:

`101000110101110`

The receiver’s check

```
                1101010110   quotient (discarded)
           ----------------
110101 ) 101000110101110    received T
         110101000000000    ⊕ P
         ----------------
          11101110101110
          11010100000000    ⊕ P shifted
          ---------------
            111010101110
                ⋮  four more steps  ⋮
                  -----
                  00000    remainder
```

The receiver gets the transmitted frame *T* and knows the divisor *P*. To check whether the frame is corrupted, it simply checks whether *T* is **divisible by *P*** — again, by long division.

Remainder = 0

In this case *T* is divisible by *P*, so the receiver declares that **no error is detected**.

One more way to write the same thing

The divisor *P* is also called the **generator** — it is what generates the FCS.

A bit string can also be written as a **polynomial**, and generators are almost always written that way.

Two reasons to use the polynomial representation:

- Every CRC **standard** publishes its generator in this form.
- It is the form in which you can **analyse** which errors a generator will catch.

Bit strings as polynomials

Each bit becomes one term: the **power** of *x* gives the bit's **position**, the **coefficient** gives its **value**. Only the 1-bits survive.

x^9	x^8	x^7	x^6	x^5	x^4	x^3	x^2	x^1	x^0
1	0	**1**	0	0	0	**1**	**1**	0	**1**

binary → polynomial

→ $x^9 + x^7 + x^3 + x^2 + 1$

Going back, write a 1 wherever a power appears and a 0 wherever it does not — here x^6, x^3, x^2 and x^1 are all missing.

$x^7 + x^5 + x^4 + 1$ →

x^7	x^6	x^5	x^4	x^3	x^2	x^1	x^0
1	0	**1**	**1**	0	0	0	**1**

polynomial → binary

The same FCS, two representations

BIT STRINGS	POLYNOMIALS
D 1010001101	$D(x) = x^9 + x^7 + x^3 + x^2 + 1$
P 110101	$P(x) = x^5 + x^4 + x^2 + 1$
F = remainder of $1010001101\mathbf{00000} \div 110101$	$F(x)$ = remainder of $D(x) \cdot \mathbf{x^5} \div P(x)$

> How do we write "append five zeros to D, then divide by P" in polynomial form?
>
> Multiply $D(x)$ by $\mathbf{x^5}$ and divide by $P(x)$ — it does exactly the same thing. Worth checking for yourself.
>
> **You do not need to learn polynomial division.** If it is unfamiliar, work in bit strings and convert the remainder at the end.

F **01110**	$F(x) = x^3 + x^2 + x$
T 1010001101**01110**	$T(x) = x^{14} + x^{12} + x^8 + x^7 + x^5 + x^3 + x^2 + x$

CRC · OPTIONAL

What can this representation tell us?

Optional material, for anyone curious about *why* the polynomial form is used. Not examined.

OBSERVATION

A corrupted frame can always be written as $\boldsymbol{T(x) + E(x)}$, where the **error polynomial** $E(x)$ has a 1 in every flipped position.

- How should we choose a good generator $P(x)$?
- We built $T(x)$ to divide evenly by $P(x)$.
- So the error slips through **exactly when $P(x)$ divides $E(x)$**.
- Therefore choose a $P(x)$ that **cannot divide** the error patterns we expect.

Example: a short burst error

Our generator is $P(x) = x^5 + x^4 + x^2 + 1$, of **degree 5**.

The **degree** of a polynomial is its highest power — equivalently, the position of its leading 1 bit.

	x^{14}	x^{13}	x^{12}	x^{11}	x^{10}	x^9	x^8	x^7	x^6	x^5	x^4	x^3	x^2	x^1	x^0
Sent T	1	0	1	0	0	0	1	1	0	1	0	1	1	1	0
Received	1	0	1	0	0	0	1	1	0	1	0	1	0	0	1
Error E	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1

E has a 1 exactly where a bit was flipped — here a burst of three

The flipped bits are the last three, so the error polynomial is $E(x) = x^2 + x + 1$ — of **degree 2**.

Example: a short burst error

Our generator is $P(x) = x^5 + x^4 + x^2 + 1$, of **degree 5**.

The **degree** of a polynomial is its highest power — equivalently, the position of its leading 1 bit.

	x^{14}	x^{13}	x^{12}	x^{11}	x^{10}	x^9	x^8	x^7	x^6	x^5	x^4	x^3	x^2	x^1	x^0
Sent T	1	0	1	0	0	0	1	1	0	1	0	1	1	1	0
Received	1	0	1	0	0	0	1	1	0	1	0	1	0	0	1
Error E	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1

E has a 1 exactly where a bit was flipped — here a burst of three

A degree-5 polynomial cannot divide a non-zero one of degree 2, so the remainder cannot be zero: **this error is caught**.

Example: a short burst error

Our generator is $P(x) = x^5 + x^4 + x^2 + 1$, of **degree 5**.

The **degree** of a polynomial is its highest power — equivalently, the position of its leading 1 bit.

	x^{14}	x^{13}	x^{12}	x^{11}	x^{10}	x^9	x^8	x^7	x^6	x^5	x^4	x^3	x^2	x^1	x^0
Sent T	1	0	1	0	0	0	1	1	0	1	0	1	1	1	0
Received	1	0	1	0	0	0	1	1	0	1	0	1	0	0	1
Error E	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1

E has a 1 exactly where a bit was flipped — here a burst of three

A more careful analysis shows this in general: **any burst shorter than *P* is always caught**, wherever it sits in the frame.

Example: a short burst error

Our generator is $P(x) = x^5 + x^4 + x^2 + 1$, of **degree 5**.

The **degree** of a polynomial is its highest power — equivalently, the position of its leading 1 bit.

	x^{14}	x^{13}	x^{12}	x^{11}	x^{10}	x^9	x^8	x^7	x^6	x^5	x^4	x^3	x^2	x^1	x^0
Sent T	1	0	1	0	0	0	1	1	0	1	0	1	1	1	0
Received	1	0	1	0	0	0	1	1	0	1	0	1	0	0	1
Error E	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1

E has a 1 exactly where a bit was flipped — here a burst of three

This is why we required *P*'s **highest-order bit to be 1** — it is what holds the degree at *r*, and the degree is what sets how long a burst can be and still be caught.

Some Standard Polynomials

These trade-offs have already been worked out. **Always use a standard generator — never invent your own.**

Four versions of *P(X)* are widely used:

- **CRC-12:** transmissions of streams of 6-bit characters
- **CRC-16 / CRC-ANSI:** various communication protocols, including serial standards like Modbus
- **CRC-CCITT:** telecommunications applications
- **CRC-32 / IEEE-802:** network protocols — Ethernet, ZIP, PNG, and other file transfer protocols

CRC-12	$= X^{12} + X^{11} + X^3 + X^2 + X + 1$
CRC-ANSI	$= X^{16} + X^{15} + X^2 + 1$
CRC-CCITT	$= X^{16} + X^{12} + X^5 + 1$
IEEE-802	$= X^{32} + X^{26} + X^{23} + X^{22} + X^{16} + X^{12} + X^{11} + X^{10} + X^8 + X^7 + X^5 + X^4 + X^2 + X + 1$

RECAP

Error detection — what we covered

- An **error-detecting code** appends check bits so the receiver can test whether the data arrived intact.
- A **parity check** adds one bit and misses any even number of flips.
- The **Internet checksum** is cheap to compute and is used by IP, TCP and UDP.
- The **CRC** appends an FCS that makes the frame divide exactly by an agreed generator — the strongest of the three.

MODULE 3 · CONCEPT 3 OF 3

Error Correction

1 · Error Detection | 2 · CRC | **3 · Error Correction**

ERROR CORRECTION

Forward Error Correction (FEC)

→ Typically, **retransmission** is needed when bit errors are detected.

→ Not good enough for **wireless applications**:

- High **bit error rate (BER)** on a wireless link
- Long **propagation delay** on satellite links

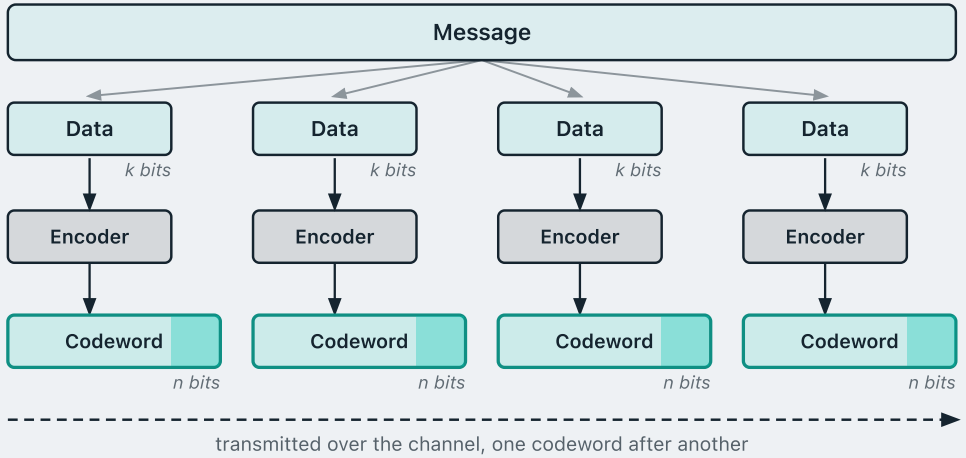
→ Need to **correct errors** on the basis of the bits received — without retransmission.

The FEC process — sender

The message is divided into blocks of k bits — each one a **data block**.

The **encoder** is applied to each data block **independently**, turning every k-bit block into an n-bit **codeword** — the extra $n-k$ bits are the redundancy.

The codewords are then transmitted, one after another.



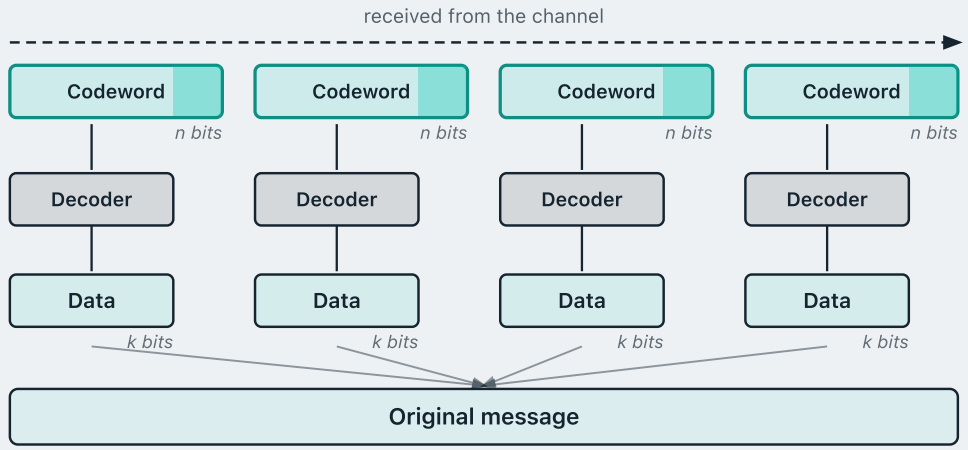
The FEC process — receiver

The receiver splits the incoming stream back into *n*-bit codewords — some of which may have been corrupted in transit.

The **decoder** is applied to each codeword independently.

Each decode recovers the original *k*-bit data block, **correcting** any errors the code is strong enough to fix.

The recovered blocks are reassembled into the original message.



Block Code Principles — Hamming Distance

The **Hamming distance** $d(v_1, v_2)$ between two n-bit binary sequences is *the number of bit positions in which v_1 and v_2 disagree*.

Example:

$v_1 =$	**0**	1	**1**	0	**1**	1
	↕	·	↕	·	↕	·
$v_2 =$	**1**	1	**0**	0	**0**	1

3 positions differ → $d(v_1, v_2) = 3$

How the encoder and decoder work

Constructing the **encoder** is the most important part — it is what determines how many errors the code can **detect** and **correct**. We will not go into those details here.

You can always think of an encoder as a **map**: a table assigning each k-bit data block one n-bit codeword. That is all we need in this lecture.

But whatever the code, the **decoder** always works in the same way.

for example, with $k = 2$ and $n = 5$:

Data block	Codeword
00	00000
01	00111
10	11001
11	11110

FEC

Minimum Hamming distance decoding

Given the received *n*-bit pattern, the decoder asks:

1. **Is it one of the valid codewords?**

   If so, **no error is detected** — output the data block the encoder assigned to it.

2. **If not, an error has occurred** — that much we know immediately.

   Then find the valid codeword **closest to it** in Hamming distance, and output *that* codeword's data block. This is **minimum-distance decoding**.

Step 2 works because **fewer bit flips are more likely than many** — the nearest valid codeword is the most plausible thing the sender actually sent.

Minimum-distance decoding — an example

Our **encoder**, with $k = 2$ and $n = 5$:

Data block	Codeword
00	00000
01	00111
10	11001
11	11110

> **Assumption:** the channel is lossy, but **at most one bit** per codeword is flipped.

The receiver gets **00100** — which is **not** one of the four valid codewords, so something was corrupted.

Which valid codeword is closest?

Received	Hamming distance	Valid codeword	Data block
00100	**1**	**00000**	**00**
00100	2	00111	01
00100	4	11001	10
00100	3	11110	11

The minimum distance is **1**, to 00000, so the decoder outputs data block **00**. Under our assumption of at most one flip, that **must** be what the transmitter sent — the error is corrected.

When can we actually correct?

Minimum-distance decoding only works if the closest valid codeword is **unique**. It need not be.

The same encoder as before:

Data block	Codeword
00	00000
01	00111
10	11001
11	11110

> **New assumption:** this channel is noisier — **up to two bits** per codeword may be flipped.

The receiver gets **01010**.

Received	Hamming distance	Valid codeword	Data block
01010	**2**	**00000**	**00**
01010	3	00111	01
01010	3	11001	10
01010	**2**	**11110**	**11**

Two codewords sit at distance 2, and two flips are now allowed. We know an error occurred — but 00 and 11 are equally plausible, so the error can be **detected but not corrected**.

The full picture for this code

With $n = 5$ there are $2^5 = 32$ possible received patterns:

`00000` `00001` `00010` … `00111` … `11001` … `11110` `11111`

Only **four** of them are valid codewords, so **32 – 4 = 28** are invalid.

- **20 of the 28** are at Hamming distance 1 from a valid codeword.
- Each is at distance 1 from **exactly one** codeword, never two — so its nearest codeword is **unique**.
- Therefore all 20 can be **corrected**. That was our first example.

`01010` `01011` `01100` `01101` `10010` `10011` `10100` `10101`

- The remaining **8** lie at distance 2 from **two different** codewords, so the nearest is **not unique**.
- So they can only be **detected** — our second example.

> So this code can **detect** any error of up to two bits, but only **single-bit errors** can be **corrected**.

What determines the ability to correct?

For a **general** code, how many errors can it correct, and how many can it detect?

It comes down to a single number: the **minimum Hamming distance** *d* of the code — the smallest distance between **any two** valid codewords.

Our code has four valid codewords:

`00000` `00111` `11001` `11110`

Pair of codewords		Distance
00000	00111	3
00000	11001	3
00000	11110	4
00111	11001	4
00111	11110	3
11001	11110	3

Every pair, six in total.

Reading off the smallest entry: **d = 3**.

FEC

Walking away from a codeword

Picture the whole space of bit strings as a **grid**: every node is one specific bit string, and any two **adjacent** nodes differ in **exactly one bit** — Hamming distance 1.

Walking away from a codeword

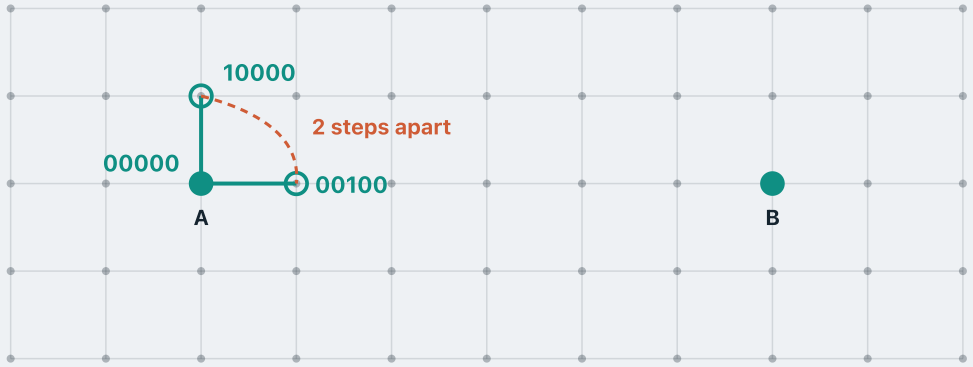
For example, if **A** is `00000`, then one step away is a string like `00100` — they differ in **exactly one bit**, so their Hamming distance is **1**.

Walking away from a codeword

Step **up** instead and we reach `10000` — a different bit flipped, but still **one step** from `00000`.

Walking away from a codeword

Notice that `10000` and `00100` are **two steps** apart — two edges to travel from one to the other.



Walking away from a codeword

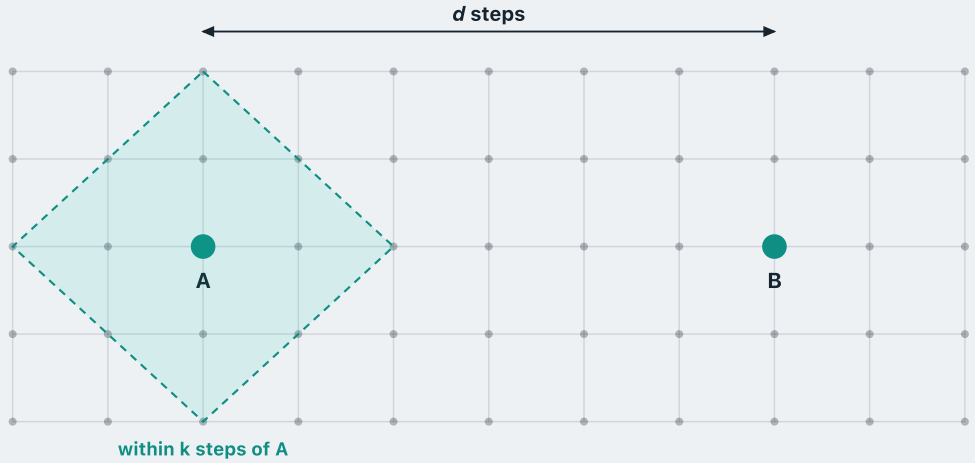
Now suppose we use a code whose **minimum Hamming distance** is d.

Walking away from a codeword

Let **A** and **B** be two codewords that achieve that minimum — on the grid they sit ***d* steps apart**.

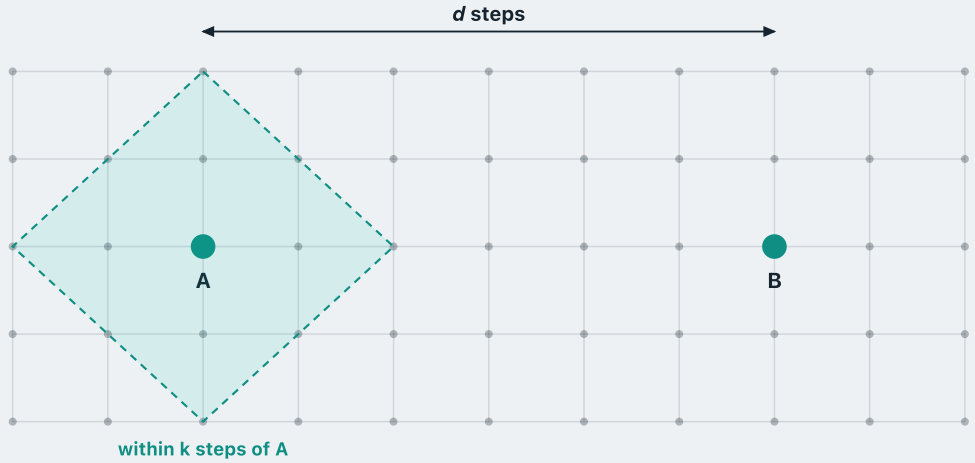
Walking away from a codeword

Send **A** over a channel that flips at most ***k*** bits. Each flip is one step to a neighbour, so the received pattern lands **somewhere within *k* steps of A**.



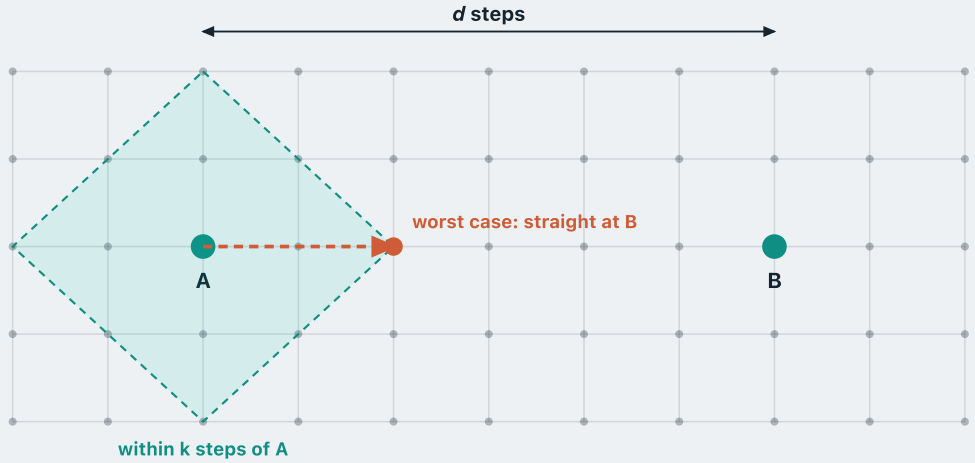
Walking away from a codeword

The further it drifts from A, the **less it looks like A** — and the more it may start to resemble some other codeword.



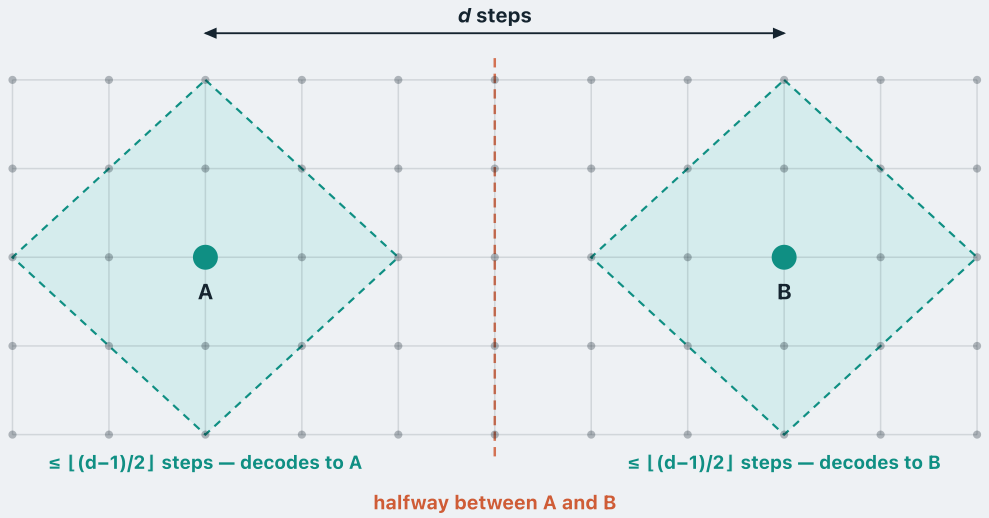
Walking away from a codeword

The **worst case** is when every flip takes us straight towards the **nearest** codeword B, so the pattern looks more and more like B with each step.



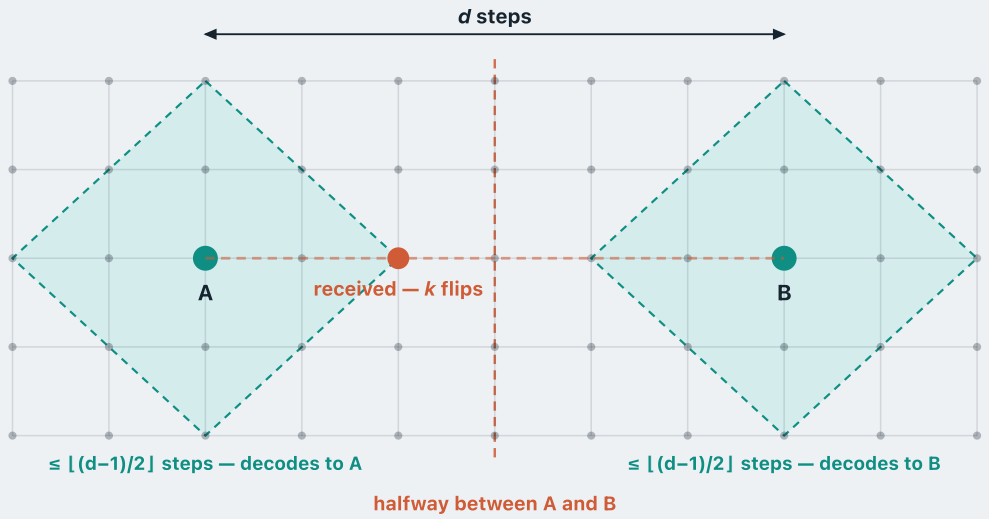
When can the error still be corrected?

Suppose the error flips k bits, with $k \le \lfloor (d-1)/2 \rfloor$. Then the received pattern stays inside **A's region** and never crosses the **halfway** line — even in the worst case.



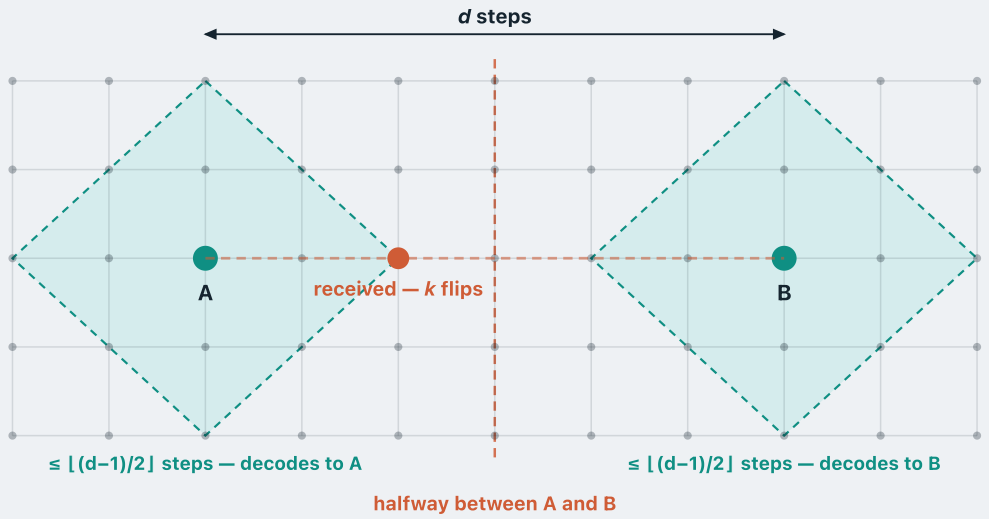
When can the error still be corrected?

What is the worst case? Every flip moves straight towards **B** — the pattern then lands *k* steps along the line from A to B, at the point shown.



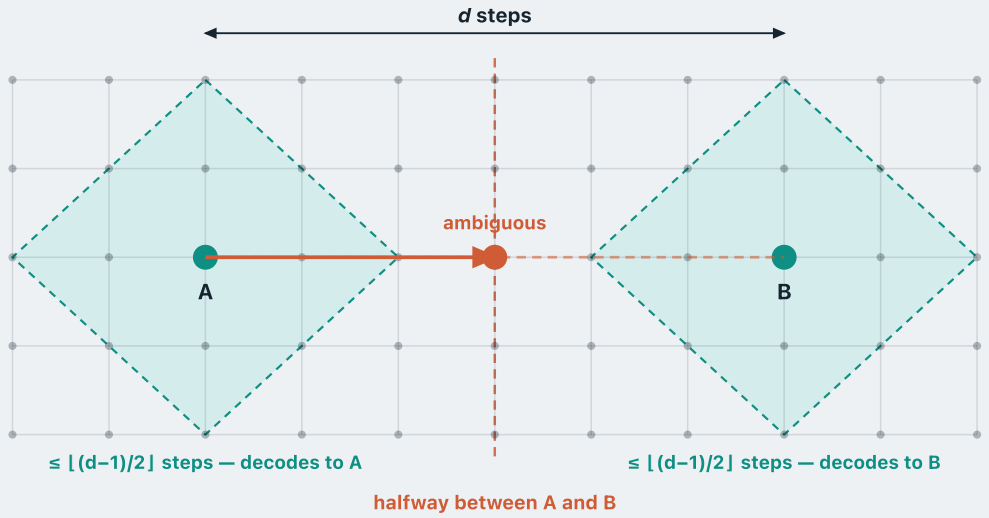
When can the error still be corrected?

So **A** is strictly the nearest codeword, and minimum-distance decoding walks us back to it: a *k*-bit error is **corrected**.



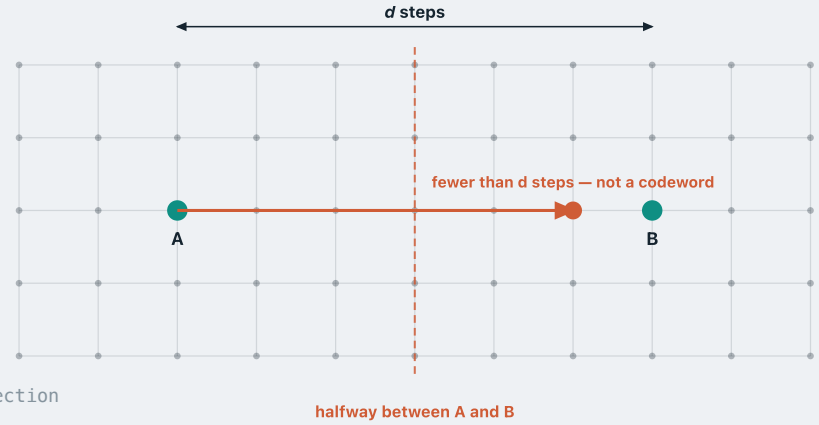
When can the error still be corrected?

Now suppose ***k* = *d*/2**. The pattern lands exactly on the **halfway** line, **equidistant** from A and B — it could have come from either, so **correction is no longer possible**.



Past halfway, we can still detect

- Suppose the channel flips **more than $\lfloor (d-1)/2 \rfloor$** bits, so correction is out of reach.
- As long as it flips **fewer than *d*** bits, we cannot reach B — B is a full *d* steps away.
- And B is the **nearest** codeword to A, so if we cannot reach B we cannot reach **any** codeword.
- So wherever the pattern lands, it **cannot be a valid codeword**.
- The receiver sees a non-codeword and reports an error — the error is **detected**, even though it cannot be corrected.



The rule

A code with minimum Hamming distance d

- **corrects** up to $\lfloor (d-1)/2 \rfloor$ bit errors
- **detects** up to $d - 1$ bit errors

For the code we have been working with, $d = 3$ — so it corrects **1** error and detects **2**.

So a good FEC code is designed to make its **minimum distance as large as possible** — the larger d is, the more errors it can correct and detect.

RECAP

Error correction — what we covered

Forward error correction repairs errors at the receiver itself, from the bits that arrived — **without retransmission**.

The **encoder** turns every k-bit data block into an n-bit **codeword**, with $n > k$; the extra $n–k$ bits are the redundancy.

The **decoder** checks whether the received pattern is a valid codeword; if it is not, an error is **detected**, and **minimum-distance decoding** returns the data block of the nearest codeword.

How many errors it can detect and correct hinges entirely on the code's **minimum Hamming distance** d.

REFERENCES

References

James F. Kurose & Keith W. Ross, *Computer Networking: a top-down approach featuring the Internet*, Addison Wesley, 3rd edition, 2005.

- Chapter 5 — The Link Layer and Local Area Network (§5.2)

William Stallings, *Data and Computer Communications*, Pearson, 10th edition, 2013.

- Chapter 6 — Error Detection and Correction

Andrew S. Tanenbaum & David J. Wetherall, *Computer Networks*, Prentice Hall, 5th edition, 2010.

- Chapter 3 — The Data Link Layer (§3.2)

ACKNOWLEDGEMENTS

Acknowledgements

These lecture notes are developed based on slides from the following sources:

- **Dr Barry Wu**'s slides for COSC264, University of Canterbury, 2024–25
- **Dr Mengmeng Ge**'s lecture notes for COSC264, University of Canterbury, 2023
- **Assoc Prof Dan Kim**'s lecture notes for COSC264, University of Canterbury, 2017
- **Prof Aleksandar Kuzmanovic**'s lecture notes for CS340, Northwestern University, 2005

Lineage & colophon

These lecture notes modernise COSC264 material developed by **Dr Barry Wu** (2024–25), building on notes by **Dr Mengmeng Ge** (2023) and **Assoc Prof Dan Kim** (2017), University of Canterbury, and lecture notes by **Prof Aleksandar Kuzmanovic** (CS340, Northwestern University, 2005).